

RANDOM * NUMBERS

Q1 Random Numbers are the samples drawn from a uniformly distributed random variables between some satisfied intervals which have equal probability of occurrence. So a random number is a number chosen as if by chance from some specified distribution such that the selection of a large set of these numbers reproduces the underlying distribution. Such numbers are always independent & uniformly distributed, so there exists no correlation between successive numbers.

Q2 Properties of Random Numbers (R.N.)

1. A sequence of Random Numbers $R_1, R_2, R_3, \dots, R_n$ must have two important statistical properties as

(i) Uniformity

(ii) Independence

2. Each R.N., R_i must be an independent sample drawn from a uniform continuous distribution betⁿ 0 and 1. The probability density function (pdf) is given by

$$\text{pdf: } f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

This pdf is shown in following figure

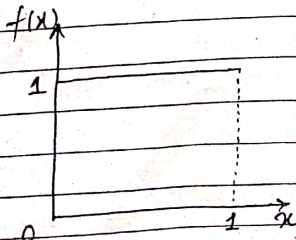


fig: The pdf for Random Number

3. The expected value of each Random Number, R_i is

$$E(R) = \int_0^1 x dx = \left[\frac{x^2}{2} \right]_0^1 = \int_0^1 x f(x) dx$$

$$\therefore E(R) = \frac{1}{2} \quad E(X)$$

and the variance is given by

$$V(R) = \int_0^1 x^2 dx - [E(R)]^2 \quad V(X) = E[X^2] - (E[X])^2$$

$$= \left[\frac{x^3}{3} \right]_0^1 - \left[\frac{1}{2} \right]^2$$

$$= \frac{1}{3} - \frac{1}{4} = \frac{4-3}{12}$$

$$\therefore V(R) = \frac{1}{12}$$

The consequences of uniformity & independence property are

- 1) If interval $[0, 1]$ is divided into n classes or the subinterval of equal length, the expected number of each observation in each interval is N/n where N = total no. of observation.

- 2) The probability of observing a value in a particular interval is independent of previous value.

Imp Pseudo Random Numbers



The random numbers generated using certain known method or arithmetic operations which reduce potential for true randomness, are called pseudo R.N. Pseudo means false, so since the method is known, the generated R.N. are not truly random and could be replicated.

Since the goal of any generation scheme is to produce a sequence of numbers betⁿ 0 & 1 that simulate or imitates, the ideal properties of uniform distribution & independence as closely as possible.

→ During generation of pseudo random numbers, certain errors or problems could occur; leading to departures from ideal randomness as following:

1. The numbers generated might not be uniformly distributed.
2. The generated numbers might be discrete values instead of continuous values.
3. The mean of generated numbers might be too high or too low.
4. The variance of generated nos. might be too high or too low.
5. There might be dependence. For Example
 - auto co-relation between numbers
 - numbers successively higher or lower than adjacent numbers
 - several numbers above mean followed by several numbers below mean.

Numerous methods can be used to generate the values. In selecting among those methods or routines, there are a number of important considerations:

1. The routine must be fast - method should be computationally efficient.

2. The routine should be portable to different computers and ideally to different programming language.

3. The routine should be sufficiently long cycle.

4. The random numbers should be applicable - completely independent of system being simulated.

5. The random numbers should closely approximate the ideal statistical properties of uniformity & independence.

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Generation of Random Numbers

In computer simulation, where a very large number of random numbers is generally required, the random numbers could be obtained by following methods:

1. Using electronic devices, very expensive
2. Using tables: R.N. might be drawn from random number tables stored in the memory of computer.
3. Using arithmetic operation.

Techniques for Generating Random Number

The linear congruential method is most widely used technique for generating random number, which was initially proposed by Lehmer in 1951.

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Linear Congruential method:

The linear congruential method or generator is an algorithm which yields a sequence of pseudo randomized numbers calculated with discontinuous piecewise linear equation.

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The linear congruential method, initially proposed by Lehmer [1951], produces a sequence of integers $x_1, x_2, \dots$ between 0 and $m-1$ by following recursive relationship:

$$X_{i+1} = (aX_i + c) \bmod m \quad \text{--- (1)}$$

where, $i = 0, 1, 2, 3, \dots$

X is a sequence of pseudo random values (integer)

X_0 , the initial value is called seed/start value

a is the multiplier, $0 < a < m$

c is the increment, $0 \leq c < m$

m is the modulus, $0 \leq X_0 \leq m$

$X, a, c, m \rightarrow$ all integer values

The above recurrence relation gives the random integers between 0 and $m-1$, where selection of a, c and m greatly determines statistical properties and cycle length.

The random number is given by

$$R_i = \frac{X_i}{m} \quad \text{where } R_i \text{ is random numbers between } 0 \text{ and } 1$$

Relation (1) can have three cases:

(1) When $c=0$, then the above generator (1) is called multiplicative congruential generator/method

i.e.

$$X_{i+1} = (aX_i) \bmod m$$

In this expression, since there is no increment, but only multiplier, a ; so the recursive relation is said to be multiplicative congruential generator.

Let us consider $X_0 = 27$, $a = 17$, ~~$b = 9$~~ , $m = 100$, $c = 0$
Then, the sequence of random integer (X_i) and
random numbers (R_i) are computed using multiplicative
method as follows:

$$X_{i+1} = (aX_i) \% m$$

$$X_1 = (17 \times 27) \% 100$$
$$= 459 \% 100 = 59$$

$$\text{and } R_1 = \frac{X_1}{m} = \frac{59}{100} = 0.59$$

$$X_2 = (17 \times 59) \% 100$$
$$= 1003 \% 100 = 3$$

$$\text{and } R_2 = \frac{X_2}{m} = \frac{3}{100} = 0.03$$

$$X_3 = (17 \times 3) \% 100$$
$$= 51 \% 100 = 51$$

$$R_3 = \frac{51}{100} = 0.51$$

$$X_4 = (17 \times 51) \% 100$$
$$= 867 \% 100$$
$$= 67$$

$$R_4 = \frac{67}{100} = 0.67$$

and so on

Hence Random integer values (X_i) = 27, 59, 3, 51, 67,

Random numbers (R_i) = 0.27, 0.59, 0.03, 0.51, 0.67, ...

(2) When $a = 1$, then the recurrence relation is called
additive congruential generator/method, i.e.

$$X_{i+1} = (X_i + c) \bmod m$$

This recurrence relation is additive congruential generator.

Example

$X_0 = 27, a = 17, c = 43, m = 100$, then

$$\begin{aligned} X_1 &= (X_0 + c) \% m \\ &= (27 + 43) \% 100 \\ &= 70 \end{aligned}$$

$$\begin{aligned} X_2 &= (X_1 + c) \% m \\ &= (70 + 43) \% 100 \\ &= 13 \end{aligned}$$

$$R_1 = \frac{X_1}{m} = \frac{70}{100} = 0.70$$

$$R_2 = \frac{13}{100} = 0.13$$

$$\begin{aligned} X_3 &= (X_2 + c) \% m \\ &= (13 + 43) \% 100 \\ &= 56 \end{aligned}$$

$$\begin{aligned} X_4 &= (X_3 + c) \% m \\ &= (56 + 43) \% 100 \\ &= 99 \end{aligned}$$

$$R_3 = \frac{56}{100} = 0.56$$

$$R_4 = \frac{99}{100} = 0.99 \text{ and so on.}$$

$$\therefore X_i = 27, 70, 13, 56, 99, \dots$$

$$R_i = 0.27, 0.70, 0.13, 0.56, 0.99, \dots$$

(3) When $c \neq 0$ and $a > 1$, then the generator is said to be mixed congruential generator.

i.e.

$$X_{i+1} = (aX_i + c) \bmod m$$

$i = 0, 1, 2, \dots$

This recursive relation gives mixed congruential generator since increment and multiplier aids in producing the random integer value.

Example

$X_0 = 27, R_0 = 0.27, a = 17, b = 43, m = 100$

then

The sequence of X_i and subsequent R_i is calculated as:

$$X_1 = (17 \times 27 + 43) \bmod 100 \quad X_2 = (17 \times 2 + 43) \bmod 100$$

$$= 502 \% 100$$

$$= 77 \% 100$$

$$= 2$$

$$= 77$$

$$R_1 = \frac{2}{100} = 0.02$$

$$R_2 = \frac{77}{100} = 0.77$$

$$X_3 = (17 \times 77 + 43) \% 100$$

$$= 1352 \% 100$$

$$= 52$$

$$R_3 = \frac{52}{100} = 0.52$$

$$\Rightarrow X_i = 27, 2, 77, 52, \dots$$

$$R_i = 0.27, 0.02, 0.77, 0.52, \dots$$

The ultimate test of linear congruential method is how closely the generated numbers $R_1, R_2, \dots$ approximate uniformity and independence. It also includes secondary properties as maximum density and period.

Q. What is Random number? What are the problems associated with generating random numbers.

Q. What are the properties of random numbers? What assumptions should we made while generating random numbers?

Q. Define pseudo R.N.? Explain the method for generating random numbers with example.

Requirements for a Good random number generator

1. The sequence of R.N. must follow the uniform distribution
2. The sequence of R.N. must be statistically independent
3. The sequence of R.N. generated must be reproducible, which allows replication of the simulation experiments.
4. The sequence must be non-repeating for any desired length.
5. Generation of R.N. must be fast because in simulation a large number of random numbers are required. A slow generator greatly increases time and cost of simulation studies/experiment.
6. The technique used in generating R.N. should require little computer memory.

TESTS FOR RANDOM NUMBERS:

The desirable properties of random numbers are uniformity & independence. Several tests can be performed in order to ensure those properties are achieved. The test could be placed in 2 categories according to properties of interest.

The first entry in the list below concerns testing for the uniformity. The second through 4th and 5th entries concern testing for independence. The five types of tests are discussed below:-

1. Frequency test: Use Kolmogorov Smirnov (K-S) test for less than 50 samples or the chi-square (χ^2) test for more (large samples) to compare the distribution of the set of numbers generated to a uniform distribution.

2. Runs test: Tests the runs up and down or the runs above and below the mean by comparing the actual values to expected values. The statistics for comparison is Chi-square (χ^2). [Independence]

8. Autocorrelation test: Tests the correlation between the numbers and compares the sample correlation to the expected correlation of zero. [Independence]

~~Imp~~ (4) Gap test: Counts the number of digits that appear between repetitions of a particular digits and then uses Kolmogorov - Smirnov test to compare with the expected size of gaps. This test is the test for independency of random numbers.

~~Imp~~ (5) Poker test: It treats numbers grouped together as a poker hand. Then the hands obtained are compared to what is expected using χ^2 -test. This test is also used for testing independence property.

* FREQUENCY TESTS - tests for uniformity i.e. U[0,1]

A basic test that should always be performed to validate a new generator is test for uniformity. They use both Kolmogorov Smirnov test and χ^2 -test.

- Both of these tests the measures, the degree of agreement between distribution of sample of generated random numbers and the theoretical uniform distribution. Both tests are based on null hypothesis of no significant differences between sample distribution & the theoretical distribution. K-S test can be applied for small sample sizes, whereas the Chi-Square is valid for large samples, say $N \geq 50$.

(1) Kolmogorov-Smirnov Test (K-S test) → Uniformity

This test compares the deviation of sample distribution with theoretical distribution, named after Andrey Kolmogorov and Nikolai Smirnov.

K-S test compares the continuous cdf, $F(x)$, of the uniform distribution with the empirical cdf, $S_N(x)$, of the sample of N observations.

By definition,

$$F(x) = x, 0 \leq x \leq 1$$

If the sample from random generator is $R_1, R_2, \dots, R_N$, then empirical cdf, $S_N(x)$, is defined by

$$S_N(x) = \frac{\text{number of } R_1, R_2, \dots, R_N, \text{ which are } \leq x}{N}$$

As N becomes larger, $S_N(x)$ should be better approxⁿ to $F(x)$, provided that null hypothesis is true.

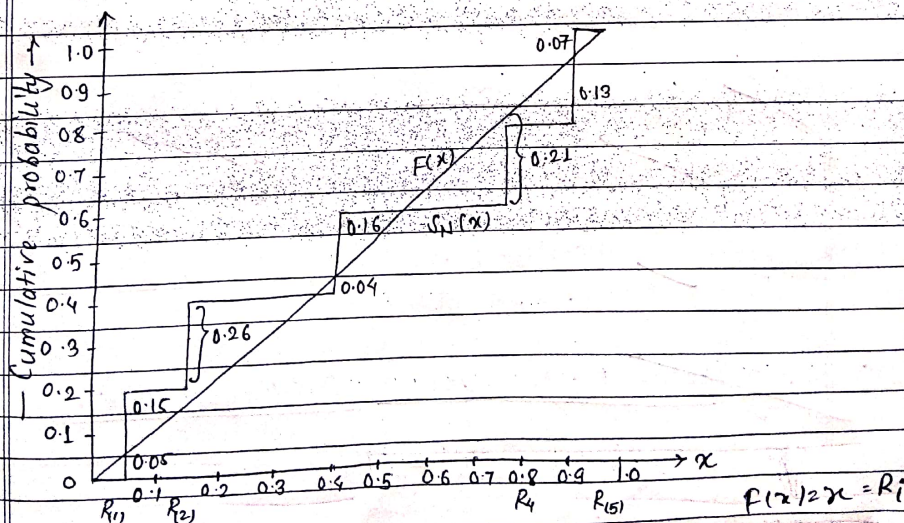


fig: Comparison of $F(x)$ and $S_N(x)$

The cdf of empirical distribution is a step function with jumps at each observed value.

The K-S test is based on largest absolute deviation between $F(x)$ and $S_N(x)$ over the range of random variable i.e. it is based on test statistics

$$D = \max |F(x) - S_N(x)|$$

D_α - critical value of D is taken from K-S Table at α , $\log_{10} N$ and N , sample observations.

ALGORITHM :

Step 1: Set up the hypothesis as

$$H_0: R_i \sim U[0,1]$$

$$H_1: R_i \not\sim U[0,1]$$

where $i = 0, 1, 2, 3, \dots$

R_i denotes i th smallest observation

Step 2: Arrange random numbers in ascending order

$$R_{(1)} \leq R_{(2)} \leq R_{(3)} \leq \dots \leq R_{(N)}$$

Step 3: Compute D^+ and D^-

$$D^+ = \max_{1 \leq i \leq N} \left\{ \frac{i}{N} - R_{(i)} \right\}$$

$$D^- = \max_{1 \leq i \leq N} \left\{ R_{(i)} - \frac{i-1}{N} \right\}$$

Step 4: Compute test statistics, D (largest difference)

$$D = \max(D^+, D^-)$$

Step 5: Determine critical value of D for given level of significance, α ; observation, N .

Step 6: Compare calculated value of D with critical value of D i.e. D_α

If $D_\alpha > D$, accept H_0 , No numbers are uniformly distributed betⁿ $[0,1]$

Else, $D_\alpha < D$, reject H_0 i.e. no evidence of uniformity has been detected

Numerical Problem

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Q. The sequence of numbers 0.63, 0.49, 0.24, 0.57, 0.71, 0.89 has been generated. Use the Kolmogorov-Smirnov test with $\alpha = 0.05$ to determine, if the hypothesis that the numbers are uniformly distributed on interval $[0,1]$ can be rejected.

Given critical value of D for $\alpha = 0.05$ & $N = 6 \Rightarrow 0.521$

Solution,

Given samples are 0.63, 0.49, 0.24, 0.57, 0.71, 0.89

1. Setting up hypothesis for uniformity as;

$H_0: R_{(i)} \sim U[0,1]$ i.e. $D_\alpha > D$

$H_1: R_{(i)} \not\sim U[0,1]$

2. Arranging above random numbers in ascending order as;

0.24 < 0.49 < 0.57 < 0.63 < 0.71 < 0.89

3. Computing D^+ & D^- from following table

i	1	2	3	4	5	6
$R_{(i)}$	0.24	0.49	0.57	0.63	0.71	0.89
i/N	$\frac{1}{6} = 0.17$	$\frac{2}{6} = 0.33$	$\frac{3}{6} = 0.5$	$\frac{4}{6} = 0.67$	$\frac{5}{6} = 0.83$	$\frac{6}{6} = 1.0$
$D^+ \quad \frac{i}{N} - R_{(i)}$	$0.17 - 0.24$ = -	$0.33 - 0.49$ = -	$0.5 - 0.57$ = -	$0.67 - 0.63$ = 0.04	$0.83 - 0.71$ = <u>0.12</u>	$1 - 0.89$ = 0.11
$D^- \quad R_{(i)} - \frac{(i-1)}{N}$	$0.24 - \frac{0}{6}$ = 0.24	$0.49 - \frac{1}{6}$ = <u>0.32</u>	$0.57 - \frac{2}{6}$ = 0.24	0.13	0.04	0.06

$$D^+ = \max \{0.04, 0.12, 0.11\}$$
$$= 0.12$$

$$D^- = \max \{0.24, 0.32, 0.24, 0.13, 0.04, 0.06\}$$
$$= 0.32$$

$$4. D = \max(D^+, D^-)$$
$$= \max(0.12, 0.32)$$
$$= 0.32$$

$$5. \text{Critical value of } D \text{ at } \alpha = 0.05, N = 6 \text{ is}$$
$$D_\alpha = 0.521 \text{ (from table i.e. K-S table)}$$

$$6. \text{Since, } D_\alpha = 0.521 > D = 0.32$$

So H_0 is accepted, the distribution is uniform.
Hence, the hypothesis that the numbers are uniformly distributed on interval $[0, 1]$ cannot be rejected.

(2) Chi-Square (χ^2) test - test for uniformity

χ^2 -test is the frequency test which measures the degree of agreement between the distribution of a sample of generated random numbers & the theoretical distribution. χ^2 -test is used for testing the uniform distribution of large sample of random numbers usually $n \geq 50$. The χ^2 -test uses the sample statistics

$$\chi_o^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

where O_i is observed number in i th class

E_i is expected number in i th class

n is number of class (class interval)

for uniform distribution, the expected value (E_i) in each class is given by

$$E_i = \frac{N}{n}$$

where N is total no. of observation

n is number of class.

It could be shown that the sampling distribution of χ_o^2 is approximately the chi-square distribution with $n-1$ degree of freedom.

Algorithm: same as K-S test with $\chi_o^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$

Q. A set of 100 random numbers have been generated as below:

0.34	0.90	0.25	0.89	0.87	0.44	0.12	0.21	0.46	0.67
0.83	0.76	0.79	0.64	0.70	0.81	0.94	0.74	0.22	0.74
0.96	0.99	0.77	0.67	0.56	0.41	0.52	0.73	0.99	0.02
0.47	0.30	0.17	0.82	0.56	0.05	0.45	0.31	0.78	0.05
0.79	0.71	0.23	0.19	0.82	0.93	0.65	0.37	0.39	0.42
0.99	0.17	0.99	0.46	0.02	0.66	0.10	0.42	0.18	0.49
0.37	0.51	0.54	0.01	0.81	0.28	0.69	0.34	0.75	0.49
0.72	0.43	0.56	0.97	0.30	0.94	0.96	0.58	0.73	0.05
0.06	0.89	0.84	0.24	0.40	0.64	0.40	0.19	0.79	0.62
0.18	0.26	0.97	0.88	0.64	0.47	0.60	0.11	0.29	0.78

Use Chi-Square test to test whether the data shown above are uniformly distributed or not. Use $\alpha = 0.05$.

Soln.

Total samples (observations) = $N = 100$

Let's divide above random number in total 10 intervals i.e. $[0 - 0.1], [0.1 - 0.2], [0.2 - 0.3], \dots, [0.9, 1]$

so $n = 10$

1. Setting up the hypothesis for testing uniformity as

$$H_0: R_{(i)} \sim U[0, 1], \chi^2_{\alpha, n-1} > \chi_0^2$$

$$H_1: R_{(i)} \not\sim U[0, 1]$$

2. Critical value of χ^2 at $\alpha = 0.05$ and

degree of freedom (df) = $n - 1 = 10 - 1 = 9$ is

Then $\chi^2_{\alpha, n-1} = \chi^2_{0.05, 9} = 16.9$

$$E_i = \frac{N}{n} = \frac{100}{10} = 10$$

n, N are chosen in such way that $n \geq 5$

Dividing above random numbers in 10 intervals from 0 to 1 as in following table with respective observed and expected frequency as:

S.N.	Interval	Observed freq (O_i)	Expected freq (E_i) = $\frac{N}{n}$	$O_i - E_i$	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
1	0-0.1	8	10	-2	4	0.4
2	0.1-0.2	8	10	-2	4	0.4
3	0.2-0.3	10	10	0	0	0.0
4	0.3-0.4	9	10	-1	1	0.1
5	0.4-0.5	12	10	2	4	0.4
6	0.5-0.6	8	10	-2	4	0.4
7	0.6-0.7	10	10	0	0	0.0
8	0.7-0.8	14	10	4	16	1.6
9	0.8-0.9	10	10	0	0	0.0
10	0.9-1.0	11	10	1	1	0.1
		100	100	$\sum \frac{(O_i - E_i)^2}{E_i} = 3.4$		

Hence from above table,

$$\chi_o^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

$$= 3.4$$

from table

$$\chi_{0.05, 10-1}^2 = 16.9$$

Since $\chi_{0.05, 9}^2 > \chi_o^2 = 3.4$, the null hypothesis of uniform distribution is not rejected i.e. accepted. Hence, the given samples of random numbers are uniformly distributed.

Gap Test (test for independence)

GAP test is the test for determining whether random numbers are independently distributed or not. This test is used to determine the significance of interval betⁿ the recurrence of same digit.

This test also compares the deviation between theoretical & observed distribution of random numbers using the same K-S statistics.

The gap length of x occurs betⁿ the recurrence of some digit.

The probability of particular gap length can be determined by Bernoulli trial. eg. let first gap length of digit 3 is 10, then

$$P(\text{gap of } n) = P(x \neq 3) P(x \neq 3) \dots P(x \neq 3) P(x = 3)$$

Since we are concerned with digits between 0 & 9, then

$$P(\text{gap of } n) = 0.9^n \cdot 0.1$$

The theoretical frequency distribution for randomly ordered digits is given by

$$P(\text{gap} \leq x) = F(x) = 0.1 \sum_{n=0}^x 0.9^n$$

$$F(x) = 1 - 0.9^{x+1} \quad \text{--- (1)}$$

The value of observed frequency distribution, $S_N(x)$ has been calculated as

$$S_N(x) = \frac{\text{Number of gaps} \leq x}{\text{Total no. of gaps}}$$

This value is equal for cumulative relative frequency for each of gap class.

ALGORITHM (GAP test)

S₁: Set up hypothesis for testing independence as

H₀: R_i is independently distributed i.e. $D_\alpha > D$

H₁: R_i is not independently

S₂: Specify the cdf for theoretical frequency distribution given by

$$F(x) = 0.1 \sum_{n=0}^x (0.9)^n$$

$$\therefore F(x) = 1 - (0.9)^{x+1}$$

S₃: Arrange the observed sample of gaps in a cumulative distribution within specified class interval, with specific interval width.

S₄: Calculate observed frequency distribution $S_N(x)$ as

$$S_N(x) = \frac{\text{number of gaps} \leq x}{\text{Total no. of gaps}} \Rightarrow \frac{\text{Frequency}}{\text{total no. of gaps}}$$

Where, Total no of gaps = total no of digits - no of distinct digits.

$S_N(x)$ is cumulative relative frequency distribution

S₅: Calculate D, maximum deviation betⁿ theoretical frequency distribution $[F(x)]$ and observed frequency distribution, $S_N(x)$; which is as;

$$D = \max |F(x) - S_N(x)|$$

S₆: Determine critical value i.e. D_α , from Kolmogorov Smirnov (K-S) table for specified value of α and sample size N .

$$\text{Use } D_\alpha = \frac{1.36}{\sqrt{N}} \text{ if } N \text{ is large}$$

N is total number of gaps.

§ Compare calculated D and critical value D_α ,
 If $D_\alpha > D$, then null hypothesis of independence is accepted, so distribution is independent

Else reject null hypothesis & distribution is not independent.

Example

Let us consider following distribution of digits (40)

4 2 (3) 6 8 9 2 1 0 4 6 8 2 (3) 1
 2 0 8 9 2 4 5 6 (3) 0 2 1 (3) 9 8
 4 6 5 2 1 0 (3) 2 5 7

Take digit 3

there are total 5 three's in above distribution

$$\therefore \text{Gap} = 5 - 1 = 4$$

first gap length of 3 = 10

$$P(\text{gap of } 10) = \underbrace{P(\text{not } 3) P(\text{not } 3) \dots P(\text{not } 3) P(3)}_{10 \text{ terms}}$$

$P(\text{not } 3) = 10 \text{ terms}$

$$\therefore F(x) = 0.1 \sum_{n=0}^{\infty} (0.9)^n = 1 - (0.9)^{x+1}$$

this is theoretical prob distribution

Here,

$$\begin{aligned} \text{Total no. of gaps} &= \text{Total no. of digits} - \text{no. of distinct digits} \\ &= 40 - 10 = 30 \end{aligned}$$

$$\text{Calculate } G_N(x) = \frac{\text{no. of gaps} \leq x}{\text{total no. of gaps}}$$

this is cumulative freq. distrib

Determine gap length of each digit

Digit	length of each gap	No of gaps
0	7, 7, 10	3
1	6, 11, 7	3
2	4, 5, 2, 3, 5, 7, 3	7
3	10, 9, 3, 8	4

⇒ Rest describe

Q. Consider following sequence of 100 digits:

2 3 5 6 0 0 1 3 4 5 6 7 9 4 9 3 1 8 3 1 3 7 4
8 6 2 5 1 6 4 4 3 3 4 2 1 5 8 7 0 8 8 2 6 7 8
1 3 5 3 8 4 0 9 0 3 0 9 2 4 6 9 9 8 5 6 0 1 7
6 7 0 3 1 0 2 4 2 0 1 1 2 6 7 6 3 7 5 9 8 6 6
7 8 2 3 5 9 6 6 4 0 3 9 3 6 8 1 5 0 7 6 2 6 0
5 7 8 0

Test whether these digits would be assumed to be independent based on frequency with which gap occurs.

Use $\alpha = 0.05$

Soln,

1. Determine the hypothesis for testing independence as

$H_0: R_{ij} \propto$ independently

$H_1: R_{ij} \neq$ independently

2. Calculate total number of gaps

$$\begin{aligned} \text{Total no of gaps (N1)} &= \text{No of digits (total)} - \text{no. of distinct digits} \\ &= 100 - 10 \\ &= 110 \end{aligned}$$

3. Determination of Gap length

Digits	length of each gap	No. of gaps
0	0, 3, 1, 12, 1, 1, 3, 4, 2, 3, 22, 7, 3, 4	13
1	9, 3, 7, 7, 10, 20, 5, 5, 26	10
2	25, 8, 7, 15, 16, 3, 12, 17	9
3	5, 7, 2, 1, 10, 0, 14, 1, 5, 16, 12, 3, 5, 6, 1	15
4	4, 8, 6, 0, 2, 17, 7, 16, 23	9
5	6, 16, 9, 11, 15, 22, 8, 11, 5	9
6	1, 6, 13, 3, 14, 16, 4, 3, 12, 1, 5, 0, 6, 8	17
	0, 5, 5, 1	



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Digits	length of each gap	No. of gaps
7	9, 16, 5, 23, 1, 12, 3, 5, 17, 5	10
8	5, 13, 2, 0, 3, 12, 29, 12, 10	10
9	1, 32, 3, 3, 0, 25, 8, 5	8

4. Cumulative Distribution Table (Using above table)

Gap length	frequency	Relative freq. = (freq/N)	Cumulative Rel. freq. $S_N(x)$	Theoretical freq. $F(x) = 1 - (0.9)^{x+1}$	$D = F(x) - S_N(x) $
0-3	34	$34/110 = 0.309$	0.309	$1 - (0.9)^{3+1} = 0.3439$	0.0349
4-7	30	$30/110 = 0.27$	0.58	$1 - (0.9)^{7+1} = 0.5695$	0.01
8-11	13	$13/110 = 0.1181$	0.70	$1 - (0.9)^{11+1} = 0.7175$	0.0175
12-15	13	$13/110 = 0.1181$	0.8146	$1 - (0.9)^{15+1} = 0.8146$	3.4×10^{-3}
16-19	9	$9/110 = 0.0818$	0.90	$1 - (0.9)^{20} = 0.8746$	0.0214
20-23	5	$5/110 = 0.045$	0.945	$1 - (0.9)^{24} = 0.92$	0.025
24-27	3	$3/110 = 0.027$	0.972	$1 - (0.9)^{28} = 0.947$	0.025
28-31	1	$1/110 = 9.09 \times 10^{-3}$	0.981	$1 - (0.9)^{32} = 0.9656$	0.015
32-35	1	9.09×10^{-3}	0.99	$1 - (0.9)^{36} = 0.977$	0.013
36-39	1	9.09×10^{-3}	1.00	$1 - (0.9)^{40} = 0.9852$	0.0148

Here, $S_N(x) = \frac{\text{No. of gaps} \leq x}{\text{Total no. of gaps}}$, $x = 3, 7, 11, 15, 19, 23, 27, 31, 35, 39$

$S_N(x)$ is cumulative relative frequency

4. Calculation of D

$$D = \max |F(x) - S_N(x)|$$

$$= 0.0349$$

5. Critical value of D at 5% LOS with $N=110$

$$D_{0.05} = \frac{1.36}{\sqrt{N}} = \frac{1.36}{\sqrt{110}} = 0.129$$

$N \rightarrow$ Total no. of gaps

6. $D_{0.05} = 0.129 > D = 0.0349 \Rightarrow \text{Accept } H_0$

So D value lies within critical region, so H_0 is not rejected. Hence given random digits are assumed to be distributed independently.

vvv

POKER TEST (test for independence)

Poker test is the test for independence based on the frequency with which certain digits are repeated with in a given series of numbers.

This test not only tests for randomness of sequence of numbers but also digits consisting of each of numbers. The expected value of each of combination of digits in a number is compared with observed value by means of χ^2 -test for independence.

Calculation of Probability

- Consider any 3 digit number, there could be only 3 possibility

- (1) All digits different
- (2) All digits same
- (3) Exactly one pair

(1) Probability that all digits are different

$$\begin{aligned} P(\text{all different}) &= P(\text{2nd digit different from 1st}) * P(\text{3rd digit diff. from 1st and 2nd}) \\ &= 0.9 * 0.8 \\ &= 0.72 \end{aligned}$$

(2) Probability that all digits same

$$P(\text{all same}) = P(\text{2nd digit same as 1st}) * P(\text{3rd digit same as 1st})$$

$$P(\text{all same}) = 0.1 \times 0.1 \\ = 0.01$$

③ Probability that ~~all~~ exactly one pair

$$P(\text{exactly 1 pair}) = {}^3C_2 \left(P(\text{2nd digit same as 1st}) \times P(\text{3rd digit diff from 1st}) \right) \\ = {}^3C_2 (0.1 \times 0.9) \\ = 0.27$$

Considering 4 Poker hand (4 digits number)
The diff possibilities are:

① $P(\text{all same}) = P(\text{2nd same as 1st}) \times P(\text{3rd same as 1st}) \times P(\text{4th same as 1st}) \\ = 1 \times 0.1 \times 0.1 \times 0.1 \\ = 0.001$

② $P(\text{all 4 digit diff}) = P(\text{2nd digit diff from 1st}) \times P(\text{3rd diff from 1st \& 2nd}) \times P(\text{4th digit diff from 1st, 2nd and 3rd}) \\ = 1 \times 0.9 \times 0.8 \times 0.7 \\ = 0.504$

③ $P(1 \text{ pair}) = {}^4C_2 (1 \times 0.1 \times 0.9 \times 0.8) \\ = \frac{4!}{2!2!} (0.1 \times 0.72) = \frac{3 \times 4}{2} (0.072) \\ = 0.432$

④ $P(2 \text{ pair}) = \frac{{}^4C_2}{2} \times \frac{{}^2C_2}{2} (1 \times 0.1 \times 0.9 \times 0.1) \\ = \left(\frac{4!}{2!2!} \right) \times \frac{1}{2} (0.01 \times 0.9) = \frac{3 \times 0.09}{2} \\ = 0.135$

$$\begin{aligned} P(3 \text{ like digits}) &= {}^4C_3 \times (1 \times 0.1 \times 0.1 \times 0.9) \\ &= 4 \times (0.01 \times 0.9) = 4 \times 0.09 \\ &= 0.036 \end{aligned}$$

Considering 5 digit numbers, there are 7 possibilities with their probabilities as:

$$\begin{aligned} (1) P(\text{All different}) &= 1 \times 0.9 \times 0.8 \times 0.7 \times 0.6 \\ &= 0.3024 \end{aligned}$$

$$\begin{aligned} (2) P(1 \text{ Pair} / 3 \text{ diff. digits}) &= {}^5C_2 (1 \times 0.1 \times 0.9 \times 0.8 \times 0.7) \\ &= {}^5C_2 (0.09 \times 0.56) \\ &= 0.5040 \end{aligned}$$

$$\begin{aligned} (3) P(2 \text{ pairs}) &= \frac{{}^5C_2}{2} \times \frac{{}^3C_2}{2} (1 \times 0.1 \times 0.9 \times 0.1 \times 0.8) \\ &= \left(\frac{5!}{2!3!} \right) \times \left(\frac{3!}{1!1!1!} \right) (0.01 \times 0.72) \\ &= \left(\frac{10}{2} \right) \times 3 (0.0072) \\ &= 0.108 \end{aligned}$$

$$\begin{aligned} (4) P(\text{three of a kind}) &= \frac{{}^5C_3}{2!1!} (1 \times 0.1 \times 0.1 \times 0.9 \times 0.8) \\ &= \frac{5!}{2!1!1!} (0.01 \times 0.72) \\ &= 0.072 \end{aligned}$$

$$\begin{aligned} (5) P(\text{Full House}) &= {}^5C_3 \times {}^2C_2 (1 \times 0.1 \times 0.1 \times 0.9 \times 0.1) \\ &= 10 \times 1 (0.001 \times 0.9) \\ &= 0.009 \end{aligned}$$

$$\begin{aligned} (6) P(\text{Four of a kind}) &= {}^5C_4 (1 \times 0.1 \times 0.1 \times 0.1 \times 0.9) \\ &= 5 (0.001 \times 0.9) = 0.0045 \end{aligned}$$

$$\begin{aligned} (7) P(\text{five of a kind/all same}) &= 1 \times 0.1 \times 0.1 \times 0.1 \times 0.1 \\ &= 0.0001 \end{aligned}$$

WVI

Numerical Problem (Poker test)

Algorithm

S₁ Define hypothesis for testing independence as

$H_0: R_i \sim \text{independently}$

$H_1: R_i \nmid$ "

S₂ Generate frequency distribution table for 3 or 4 or 5 comb^s and apply χ^2 -test.

S₃ Compute test statistics

$$\chi^2_o = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

S₄ Determine critical value at given LOS, α and $df = n-1$

S₅ If $\chi^2_o > \chi^2_{\alpha, n-1} \Rightarrow H_0$ is rejected

Else H_0 is accepted

Q. A sequence of 10,000 five digit numbers has been generated and an analysis indicates following combination with frequencies

Combination (P)	Observed frequency (O _i)
All different	3450
One pair	4510
Two pair	1150
Three of a kind	750
Full house	85
Four of a kind	40
Five of a kind	15

Based on Poker test, check whether the numbers are independent. χ^2 at $\alpha=0.05$ & $df=6$ is 12.592

Soln.

Setting up the hypothesis for testing independence as

H_0 : R_i independently

H_1 : R_i is independently

Since there are 5 different digits i.e. 5 poker hand, then

$$(1) P(5 \text{ different digit}) = 1 \times 0.9 \times 0.8 \times 0.7 \times 0.6 \\ = 0.3024$$

$$(2) P(5 \text{ same digit}) = 1 \times 0.1 \times 0.1 \times 0.1 \times 0.1 \\ = 0.0001$$

$$(3) P(\text{one pair}) = {}^5C_2 (1 \times 0.1 \times 0.8 \times 0.8 \times 0.7) \\ = 0.5040$$

$$(4) P(\text{two pair}) = {}^5C_2 \times {}^3C_2 (1 \times 0.1 \times 0.9 \times 0.1 \times 0.8) \\ = 5 \times 3 (0.01 \times 0.72) \\ = 0.108$$

$$(5) P(\text{Three of a kind}) = {}^5C_3 (1 \times 0.1 \times 0.1 \times 0.9 \times 0.8) \\ = 10 (0.01 \times 0.72) \\ = 0.072$$

$$(6) P(\text{Four of a kind}) = {}^5C_4 (1 \times 0.1 \times 0.1 \times 0.1 \times 0.9) \\ = 5 (0.001 \times 0.9) \\ = 0.0045$$

$$(7) P(\text{full house}) = {}^5C_3 \times {}^2C_2 (1 \times 0.1 \times 0.1 \times 0.9 \times 0.1) \\ = 10 \times 1 (0.001 \times 0.9) \\ = 0.009$$

The expected value for each combination is calculated by following expression

Expected frequency (E_i) = Probability $\times$ N

Here, $N = 10,000$

eg. for all different digits

$$E_1 (\text{for all diff}) = P \times N \\ = 0.3024 \times 10,000 \\ = 3024$$

$$E_2 (\text{one pair}) = 0.5040 \times 10,000 \\ = 5040$$

S.N.	Combinations (i)	Observed freq. (O_i)	Expected freq. ($E_i = P_{rob} \times N$)	($O_i - E_i$)	($O_i - E_i$) ² E_i
1	5 different digits	3450	$3024 \times 10,000 = 3024$	426	180.01
2	One pair	4510	5040	-530	526.00 55.75
3	Two pairs	1150	1080	70	4.537
4	Three of a kind	750	720	30	1.25
5	Full house	85	90	-5	0.278
6	Four of a kind	40	45	-5	0.556
7	Five of a kind	15	1	14	196

$$\sum \frac{(O_i - E_i)^2}{E_i} = 318.38$$

from above distribution table

$$\chi_o^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} = 60.01 + 55.75 + 4.537 + 1.25 + 0.278 + 0.556 + 196 = 318.38$$

Now, $df = n - 1 = 7 - 1 = 6$, $\alpha = 5\% = 0.05$

from χ^2 table

$$\chi_{0.05, 6}^2 = 12.592$$

$\therefore \chi_{0.05, 6}^2 < \chi_o^2$, so H_0 is rejected

Hence, the given numbers are not independent and there is difference betn observed and expected distributions.

Generating discrete distributions (Self study)